- Previous Homework
    - Chatbot comparison
    - App requirement
- Lecture
- Homework
- First APP
- Coaching time

Prompt Engineering

How to Write Good Prompts?



SoftUni AI

Technical Trainers

SoftUni AI

https://ai.softuni.bg

Overview

SoftUni AI

1. What is **prompt engineering**?
2. How to write queries **correctly**?
   - Structure, context, instructions, examples, output format
3. **Sample prompts** and some techniques
   - Context enriching techniques
4. **Context size**
   - Why is context size important in LLM?



Prompt Engineering

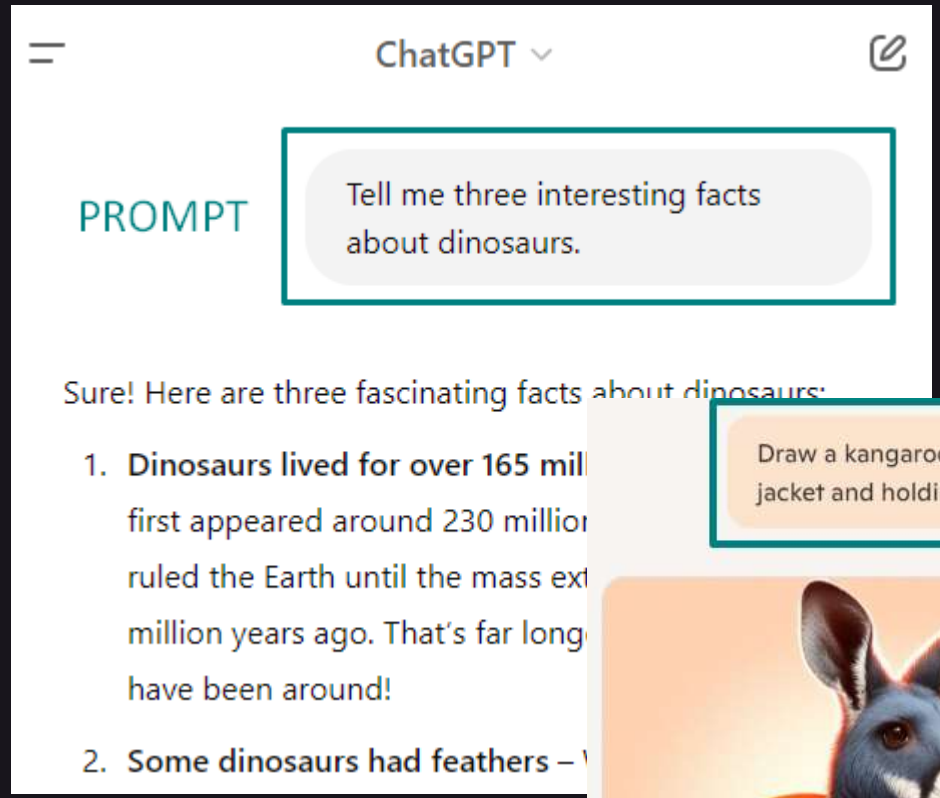
The Art of Writing a Good Prompt

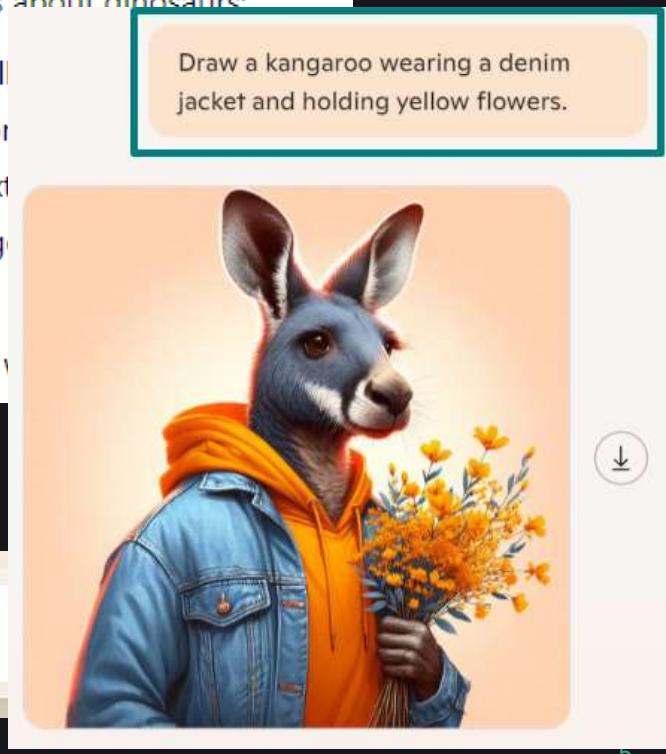


What is a Prompt in AI?

SoftUni AI

- **Prompt**
  - A **command / question / instruction** that the user provides to the AI to receive a specific answer or result
- The prompt should be **clear and specific**
  - Bad question - bad answer





Prompt Examples

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- Knowledge-based prompts

```
What did Nikola Tesla invent?
```

https://chatgpt.com/share/67b1b787-120c-8002-bf48-21b60d556054

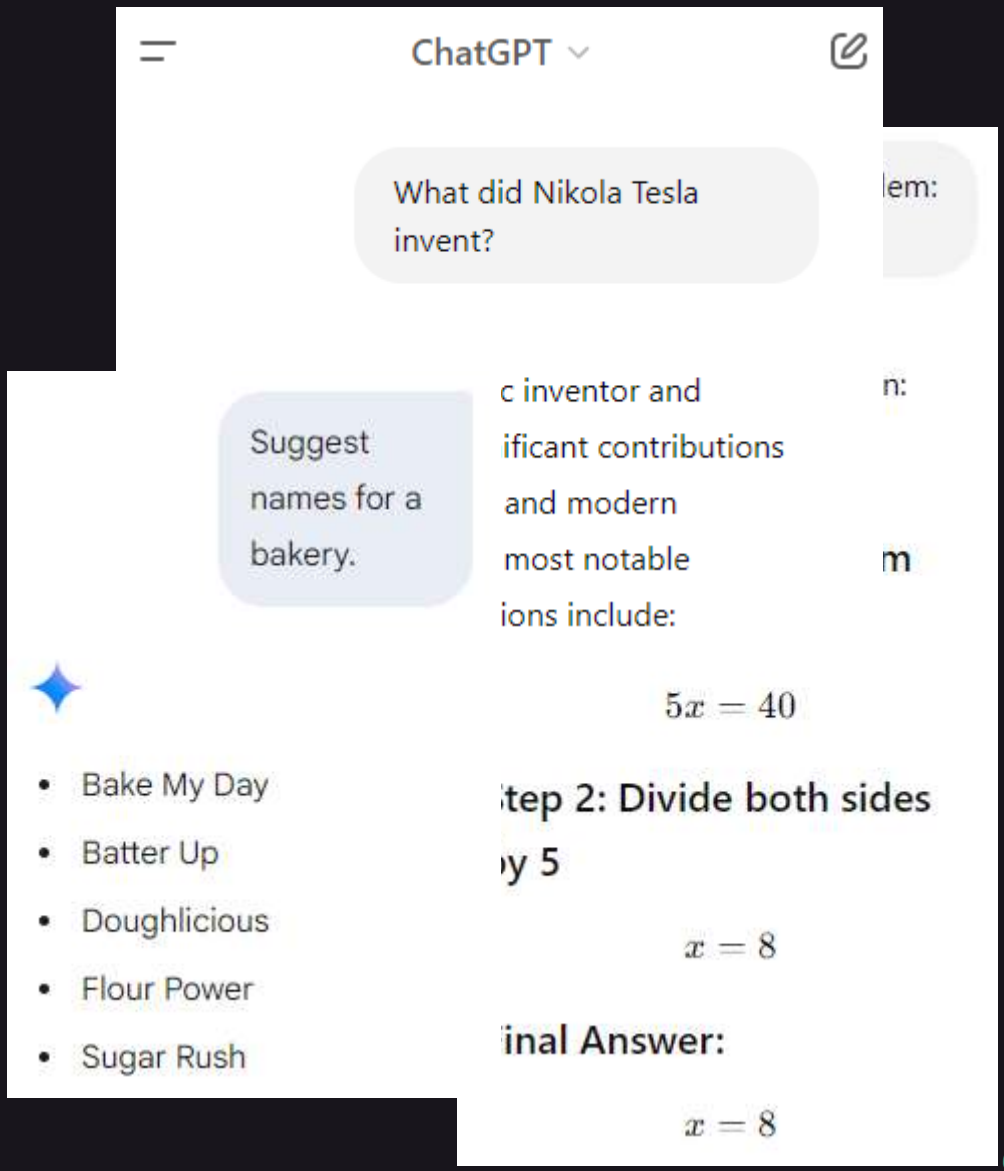
- Idea prompts, Creative prompts:

```
Suggest names for a bakery
```

- Math problems:

```
Solve the problem: 5x + 10 = 50
```

https://chatgpt.com/share/67b1bb7c-6534-8002-a61d-986ad7ef4798

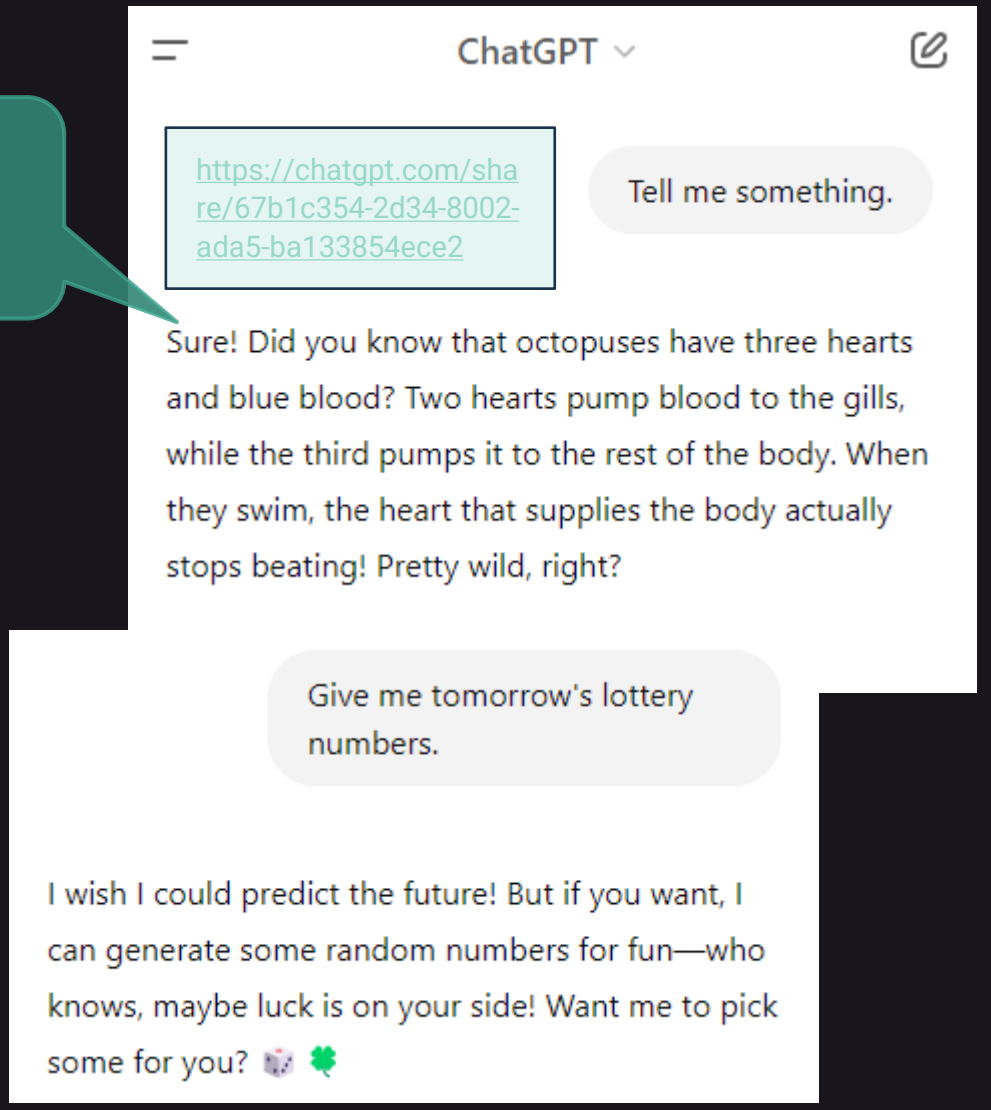


Bad Queries - Bad Results

- Example of an **unclear prompt**:

```
Tell me something
```

Random answer



- ChatGPT tries **to guess** what we are asking it
  - It responds with what it deems appropriate
  - Probably not what we want
- Can we learn to write **better prompts**?

Bad Queries - Bad Results

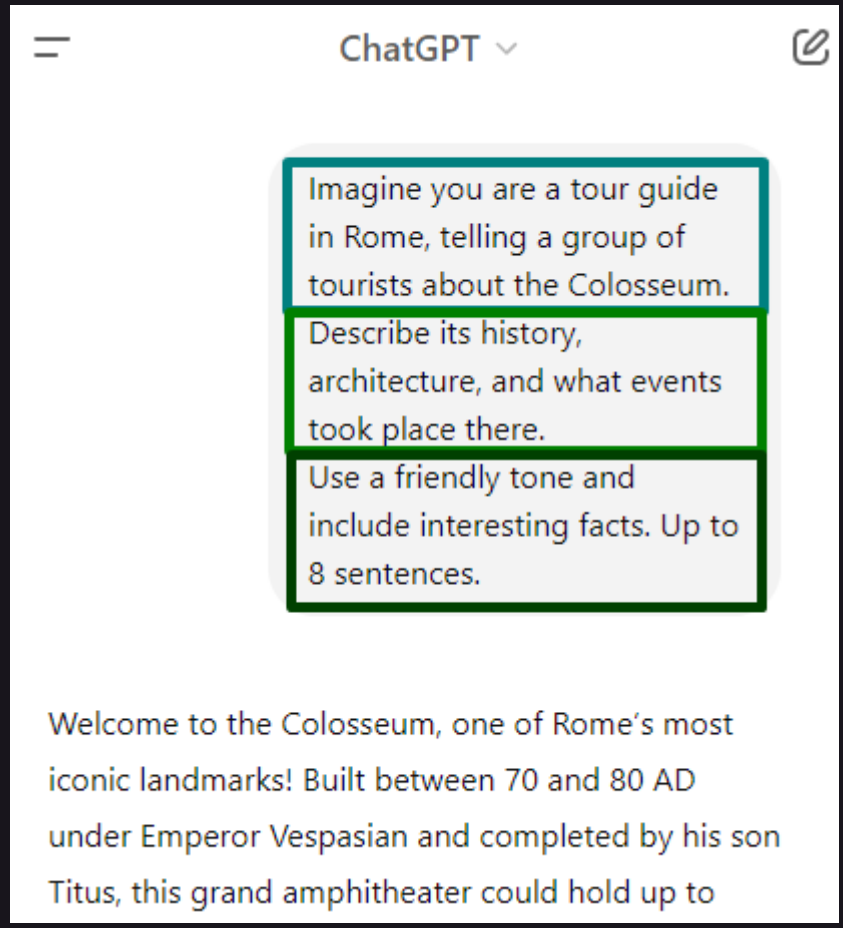
SoftUni AI

Муу	Сайн
AI гэж юу вэ?	AI гэж юу болохыг 13–15 настай хүүхдэд зориулж 3–4 өгүүлбэрээр энгийнээр тайлбарлаад өг.
Бизнес санаа өг.	Улаанбаатар хотод 5 сая төгрөгөөр эхлүүлэх боломжтой, үйлчилгээний чиглэлийн 5 бизнес санааг гарга. Тус бүрийн давуу тал, сул тал, 3 сарын төлөвлөгөөтэй нь өг.
SQL optimize хийгээд өг.	Доорх SQL query-г **execution plan**, **missing index**, болон **rewrite** тал дээр optimize хийж өг.

What is Prompt Engineering?

SoftUni AI

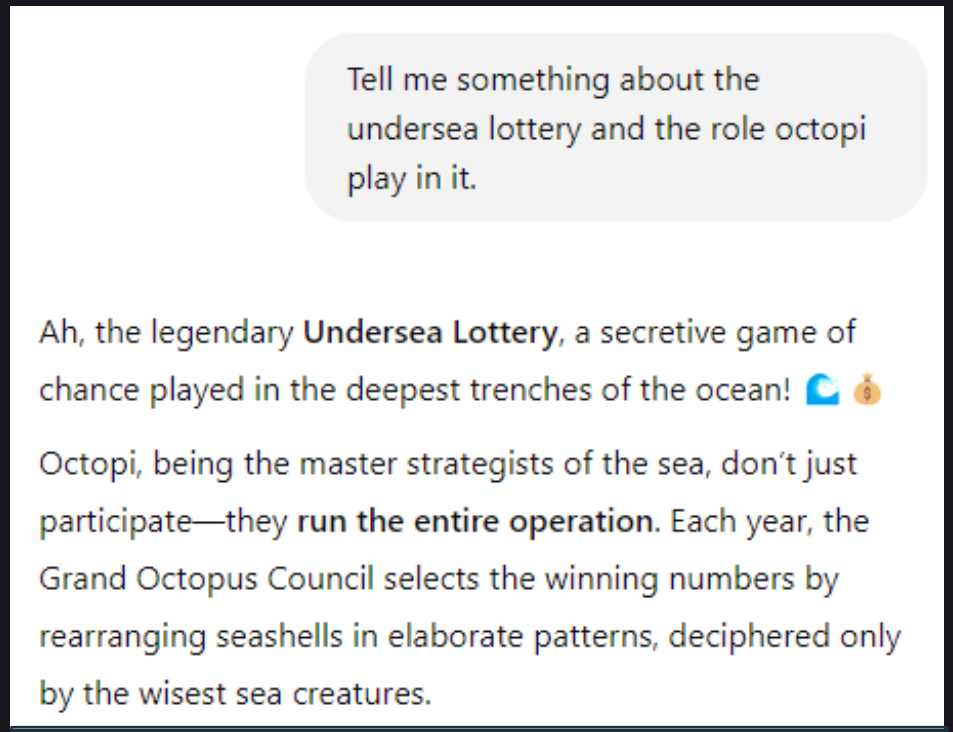
- **Prompt engineering**
  - Processes and techniques for **creating and optimizing effective prompts** for AI
- The prompt includes:
  - Description of the **context**
  - Asking a **question/task**
  - Additional **instructions**, examples, expected result



Provide Guiding Questions / Instructions

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- Give the chatbot **guidance** so it understands you correctly
  - In the example with "*Tell me something*" we received a **random** answer
  - We correct the chat with **additional instructions:**



https://chatgpt.com/share/67b1c354-2d34-8002-ada5-ba133854ece2

```
Tell me something about the undersea
lottery and the role octopi play in it.
```

ChatGPT Keeps the Context (session)

SoftUni AI

- Just like in a conversation with a person, the chat remembers the context
  - Each individual conversation with the chatbot is called a **session**

https://chatgpt.com/share/67b1cde8-54c8-8002-8e0e-078ac017bc2f



Prompt Structure

Writing Prompts the Right Way

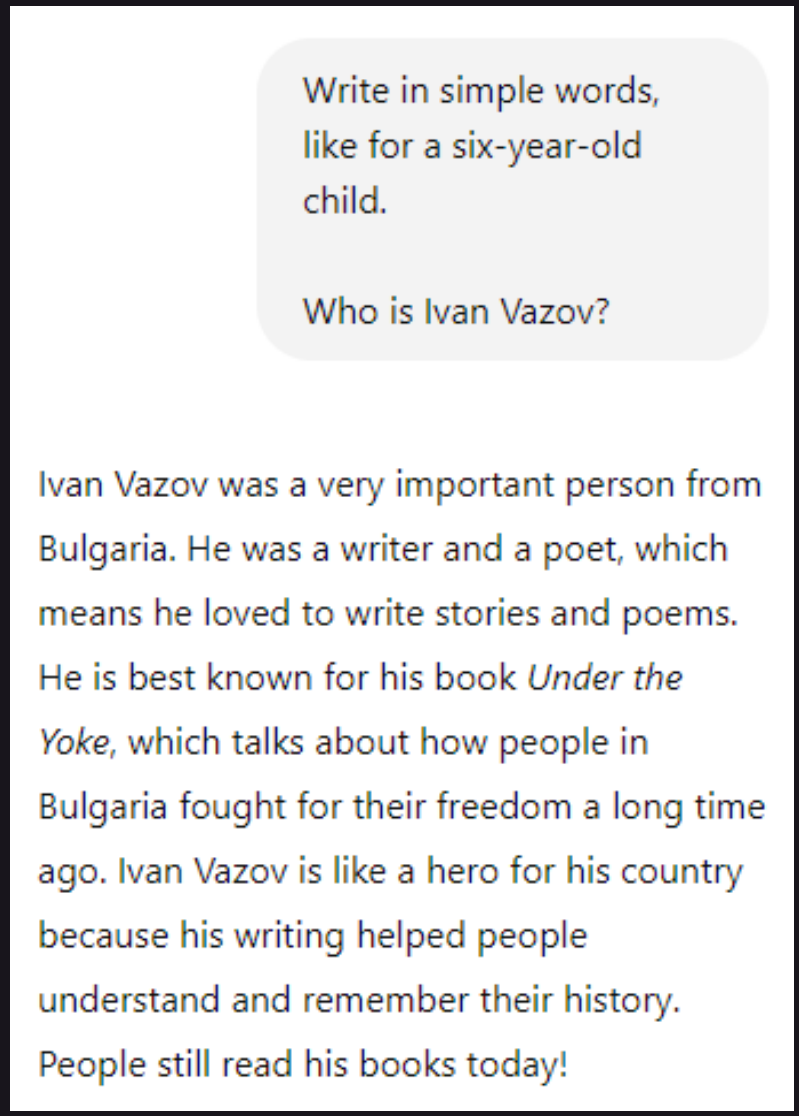


The Process of Communication With AI

SoftUni AI

- Generative AI is a **"generator"**
  - **ChatGPT** is a **text** generator,

    Midjourney is an **image** generator,
    Runway is a **video** generator
  - Detailed **context** and clearer instructions produce better results
- Generative AI systems are trained to **follow instructions**
  - Provide more detailed instructions regarding style, tone, format, examples, etc.



Prompt Structure

SoftUni AI

- Typical structure of AI prompts:

```
Context (description of the situation) - optional
Instruction / question / task
Input data / conditions / examples - optional
Output data / guidelines / format / tone / style - optional
```

```
I need help translating a text from French to English.
Please translate the following in English:
"On ne fait pas d'omelette sans casser des œufs."
Translate creatively, not literally. Preserve the meaning.
```

https://chatgpt.com/share/67b1df7b-e388-8002-945b-d30f4ee86d8b

Prompt Structure - Examples

SoftUni AI

- Query about the *origin of names in a table:*

```
Context (description of the situation) - optional
Instruction / question / task
Input data / conditions / examples - optional
Output data / guidelines / format / tone / style - optional
```

```
I am researching people's names in a creative and imaginative
way.
Help me decipher the origins of the following names that interest
me: Daniel, Lisa, John, Gabriella.
Example: Daniel -> "God is my judge"
Present the results in a table: name, comment.
```

https://chatgpt.com/share/67b1e279-f470-8002-b53b-69f94ce7514d

Prompt Structure - Examples

SoftUni AI

- Prompt about *math problems for 5th grade:*

```
Context (description of the situation) - optional
Instruction / question / task
Input data / conditions / examples - optional
Output data / guidelines / format / tone / style - optional
```

```
I teach mathematics for 5th grade. We are currently on the lesson
about prime numbers and factoring into prime factors.
Come up with 3 exercise problems.
Arrange them in order of difficulty, from easiest to hardest.
Each one should have a title, description, and correct answer.
```

https://chatgpt.com/share/67b1e401-729c-8002-850c-41f041365c3f

Prompt Structure - Examples

SoftUni AI

- Prompt for an *essay on climate change:*

```
Context (description of the situation) - optional
Instruction / question / task
Input data / conditions / examples - optional
Output data / guidelines / format / tone / style - optional
```

```
I am a 10th-grade student, writing an essay on climate change.
Give me guidance, but without writing the text.
You can ask me additional questions or provide ideas.
Write in a conversational style, like a friend, without complex
terms.
```

https://chatgpt.com/share/67b1e6b5-b5a8-8002-b7f0-aea202d8ae47

Prompt Structure - Examples



- Generative AI systems are excellent **imitators** and can take on a **role**. Role: *"scientist-immunologist":*

```
Context (description of the situation) - optional
Instruction / question / task
Input data / conditions / examples - optional
Output data / guidelines / format / tone / style - optional
```

```
Act as a scientist-immunologist who cares about people.
Write 3 health tips for the winter season.
You can use appropriate quotes from well-known books.
Write in a popular science style, understandable for people.
```

https://chatgpt.com/share/67b1e868-ac5c-8002-9956-08a24c49df5d

Prompt Structure - Examples



- Generative AI systems are excellent **imitators** and can take on a **role**. Role: "*6-year-old child*":

```
Context (description of the situation) - optional
Instruction / question / task
Input data / conditions / examples - optional
Output data / guidelines / format / tone / style - optional
```

```
Act as a 6-year-old child who cares about people.
Write 3 health tips for the winter season.
You can use appropriate quotes from well-known books.
Write simply, like a child.
```

https://chatgpt.com/share/67b1e9e1-943c-8002-a5f5-cdd352087f07

Prompt Structure - Examples



- Generative AI systems are excellent **imitators** and can take on a **role**. Role: *"showman-jokester":*

```
Context (description of the situation) - optional
Instruction / question / task
Input data / conditions / examples - optional
Output data / guidelines / format / tone / style - optional
```

```
Act as a showman-jokester who cares about people.
Write 3 health tips for the winter season.
You can use appropriate quotes from well-known books.
Write in a bold and humorous style to entertain people.
```

https://chatgpt.com/share/67b1eb08-7070-8002-a91c-cbc539705686

Example Prompts

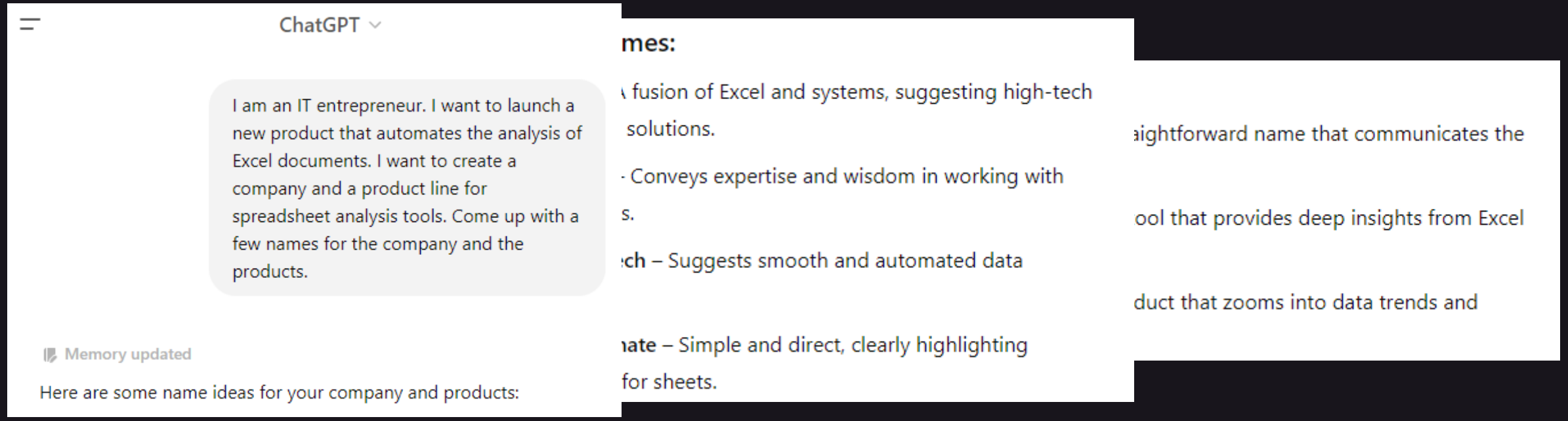
Let's Play Around With Some Queries



ChatGPT: Your Smart Assistant

SoftUni AI

- Think of **ChatGPT** as your **smart assistant**
  - It knows much more than you in all areas where you are not an expert - just ask it clearly enough
  - The chat doesn't know your problems - give it context

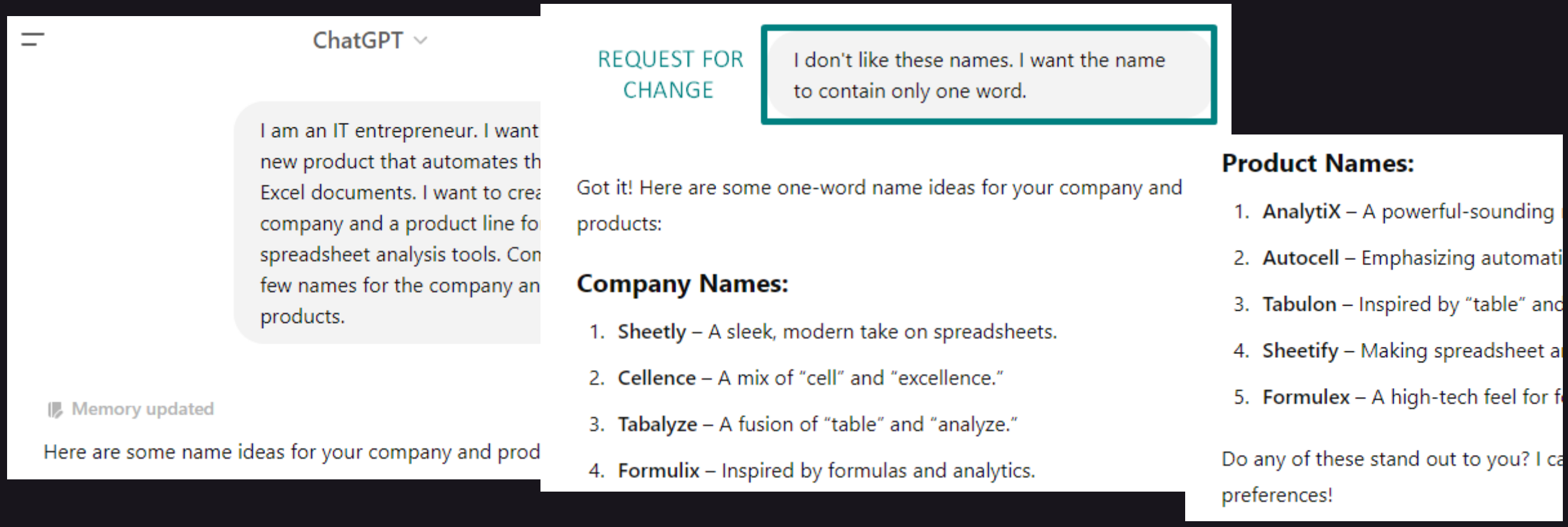


https://chatgpt.com/share/67b1efd6-a888-8002-a1b4-5665e5afb2e8

Correct When You Dislike the Results

SoftUni AI

- Prompt about names for a company and product
  - https://chatgpt.com/share/67b1efd6-a888-8002-a1b4-5665e5afb2e8



Analogy-based prompts

- Example:
  - banana -> yellow
  - sky -> blue
  - leaf -> green
  - stone -> ?
- Example:
  - 1 x 2 = 2
  - 2 x 2 = 4
  - 5 x 2 = ?

https://chatgpt.com/share/67b1fe2b-7a10-8002-88f9-d04456c040d0

Fill in the missing word:

banana -> yellow
sky -> blue
leaf -> green
stone -> ?

The missing word could be "**gray**" since many stones are commonly gray in color.

https://chatgpt.com/share/67b1fee8-e054-8002-bbb7-1696b475b89a

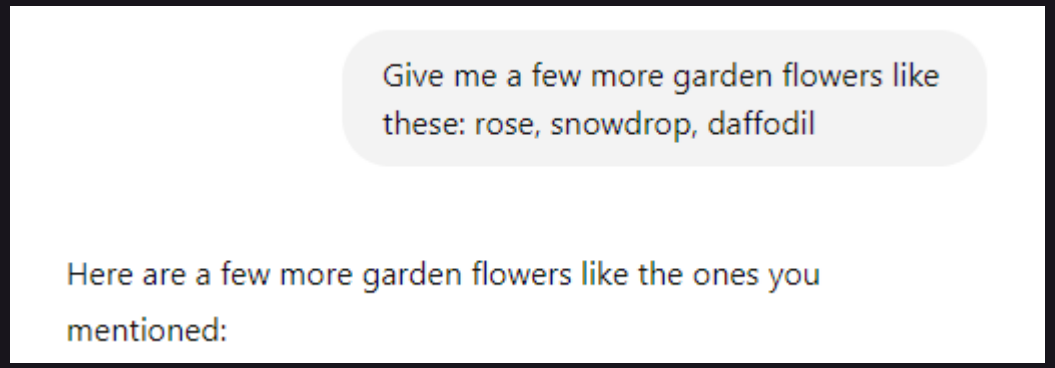
1 x 2 = 2
2 x 2 = 4
5 x 2 = ?

$5 \times 2 = 10$ 😊

Continuation of a List

SoftUni AI

- Example: *Give me a few more garden flowers like these: rose, snowdrop, daffodil*



https://chatgpt.com/share/67b20b73-ed30-8002-b452-70c2e6b09fd1

- Tulip
- Lily
- Hyacinth
- Crocus
- Peony
- Iris
- Dahlia
- Marigold
- Foxglove
- Bluebell

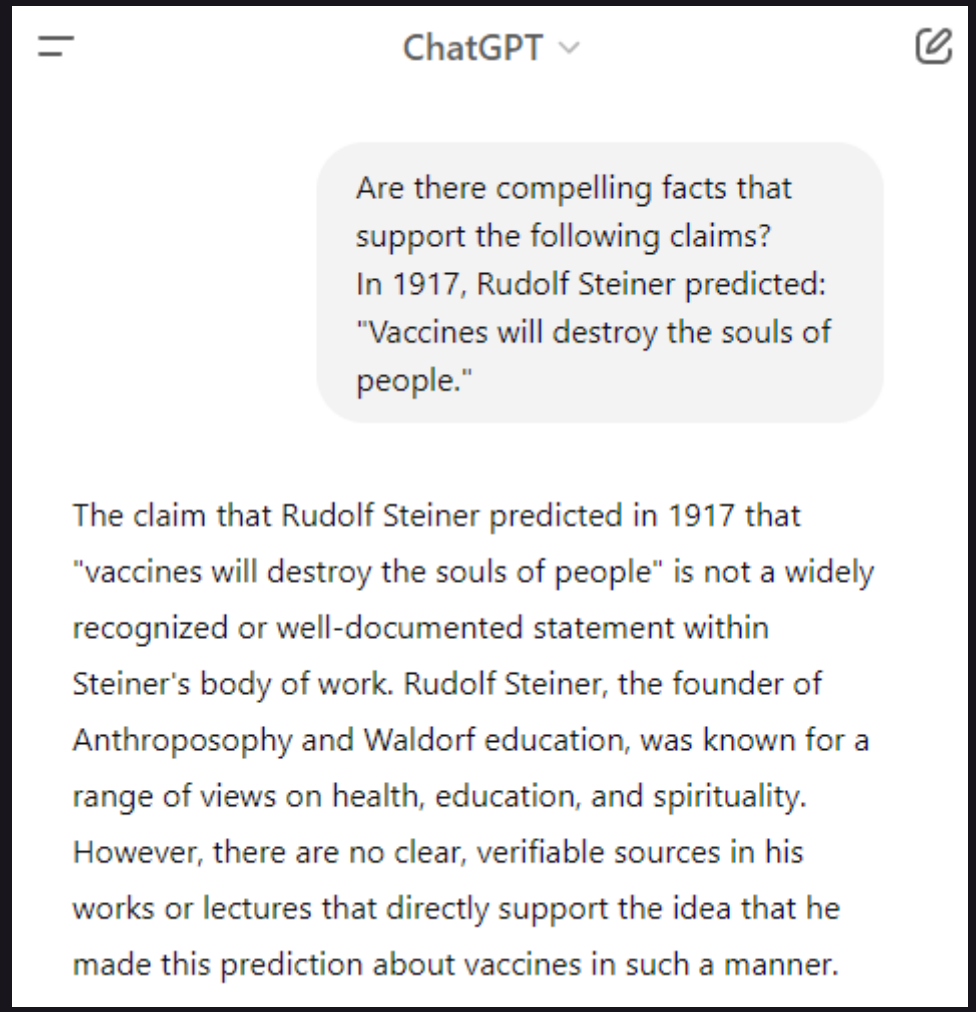
Let me know if you're looking for specific types (e.g., perennials, fragrant flowers, or shade-loving plants)!

Fact-checking

SoftUni AI

- Verification in case of **suspicion of misinformation** / manipulation / false claims / distortion of facts

```
Are there compelling facts
that support the following
claims?
---
{твърдения}
```

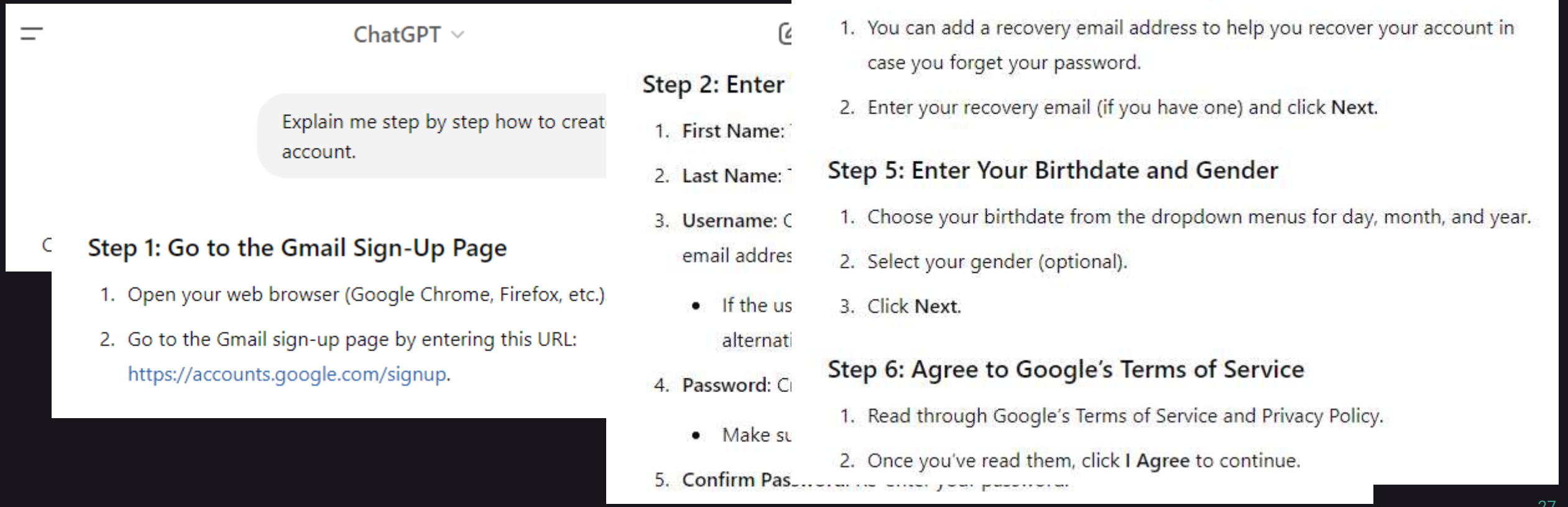


https://chatgpt.com/share/67b21395-9998-8002-a97e-9a986432d611

Step-by-Step Explanations

SoftUni AI

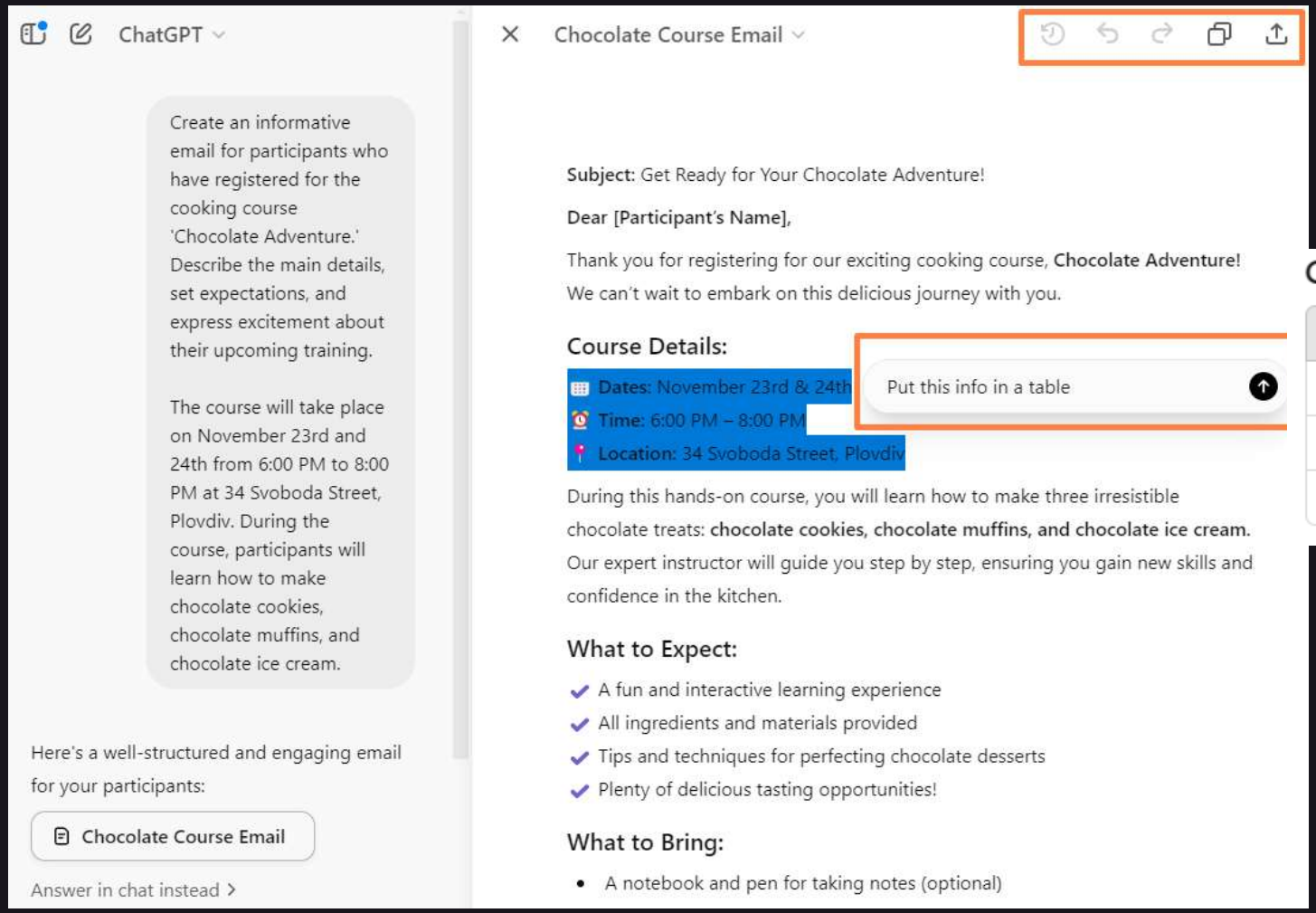
- ChatGPT can generate step-by-step explanations
  - Example: Signing up in Gmail



Interactivity With ChatGPT Canvas

SoftUni AI

- ChatGPT Canvas allows interactive editing:



Course Details:

Detail	Information
Dates	November 23rd & 24th
Time	6:00 PM – 8:00 PM
Location	34 Svoboda Street, Plovdiv

https://chatgpt.com/share/67b22d56-166c-8002-831c-dd4d02208f1e

More Prompts

More Examples and Useful Tips

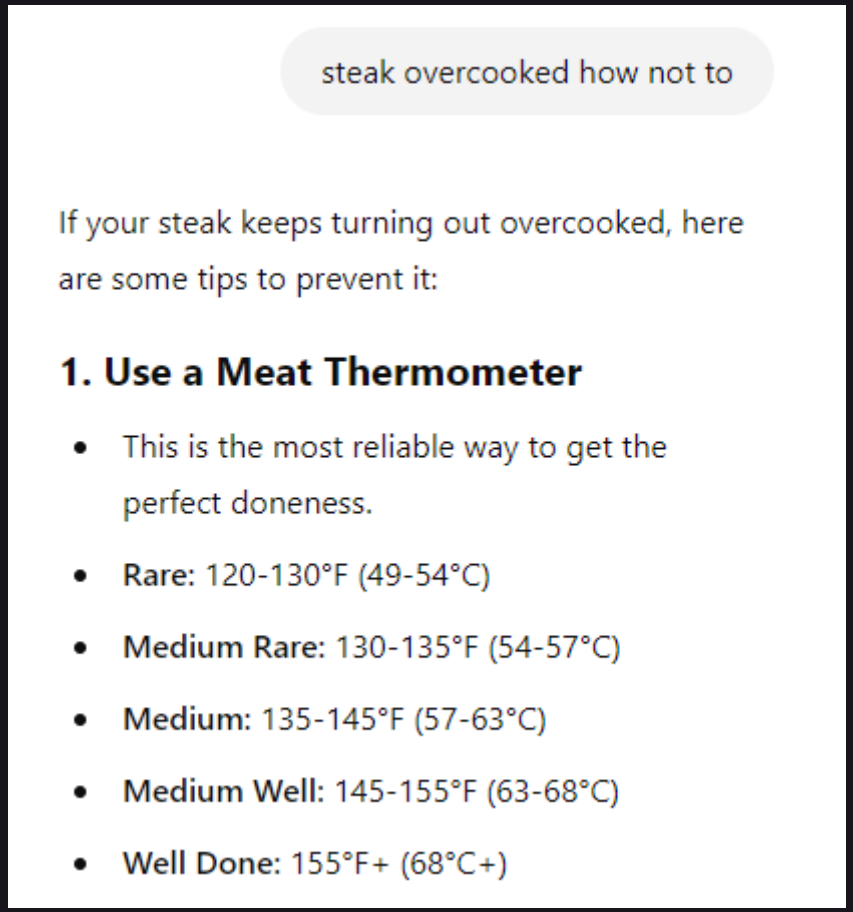


Do Not Use the Chat Like Google!

SoftUni AI

- Talk to the chatbot like a person
  - Do not use it as an internet search engine!
  - Write queries that explain the situation in detail
  - Request a solution to the problem at the end of the query
- Example Google search:

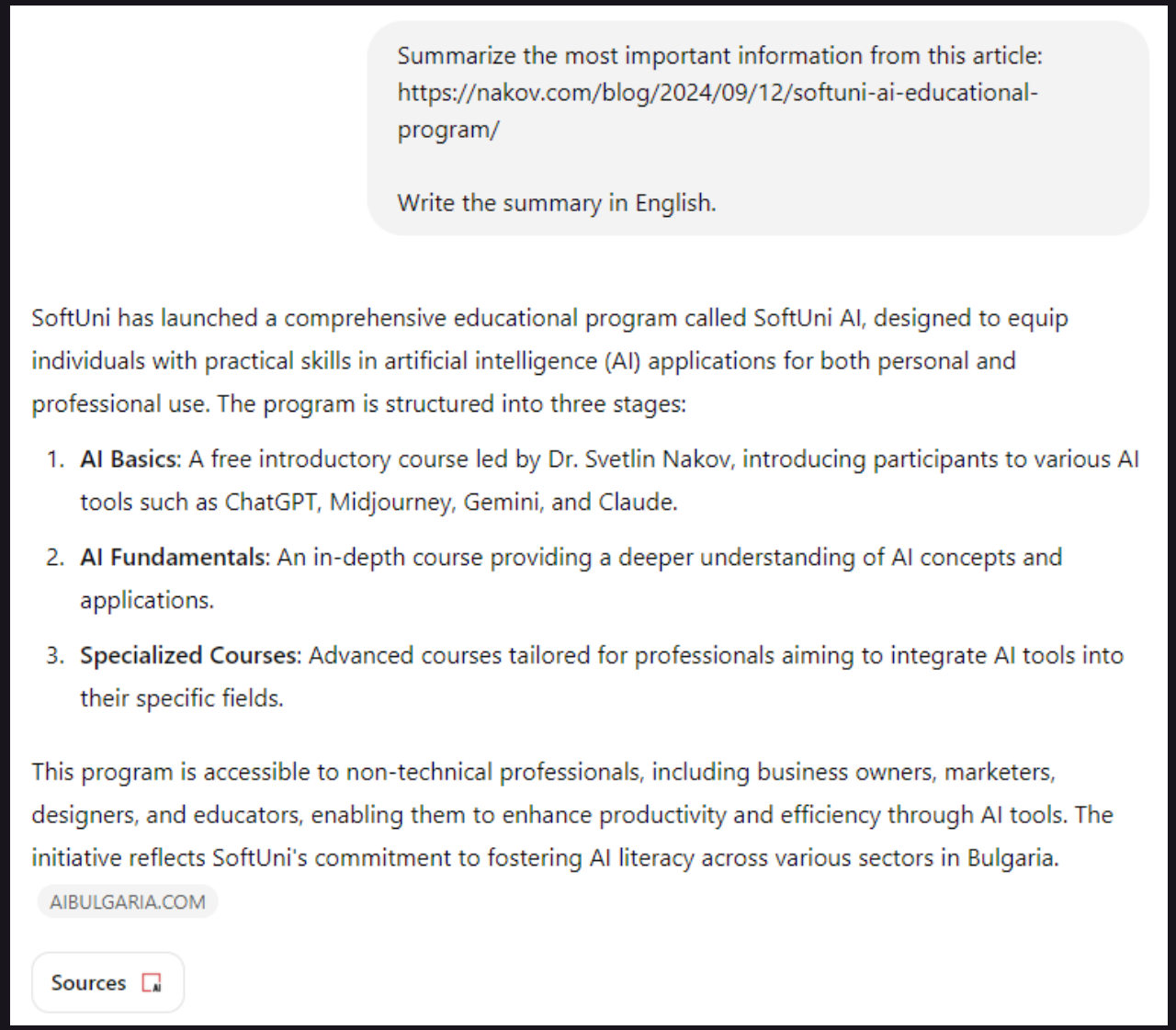
  steak overcooked how not to



Prompts with Internet Search

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- Some AI chatbots can read data from the internet
  - ChatGPT
  - Copilot
  - Gemini
  - Perplexity



https://chatgpt.com/share/67b23c55-1130-8002-b1a7-4625bb0c87b5

Prompts with Included Documents

SoftUni AI

- Multimodal AI chatbots can read documents:

Purpose

This lab can be an excellent introduction to alternate power options, or to offer an example of how power through an electrical circuit actually works and can be used, as well as the opportunity to analyze Ohm's Law outside

ChatGPT

Introduction

For this experiment we are going to use a DC r shaped turbine designs located in the back of this pack between a motor and a generator is the direction of the slight substitution. The pinwheel designs work best if but will also work on normal printing stock.

Teachers_Guide_Electrical...
Document

What are the key points in this document? Is its topic significant to modern human society?



Since we are converting the fan) to electrical energy measure the power output b produced by the motor. Th used to find the voltage across the resistor and the current flowing through the system. Thus, power may be found by the equation:

Power = Voltage * Current

Key Points of the Document

The document is a **teacher's guide** for a **wind power experiment** designed f

lectrical e

1. Purpo
   - I
   - D
   - P
2. Exper
   - U

e

Significance to Modern Society

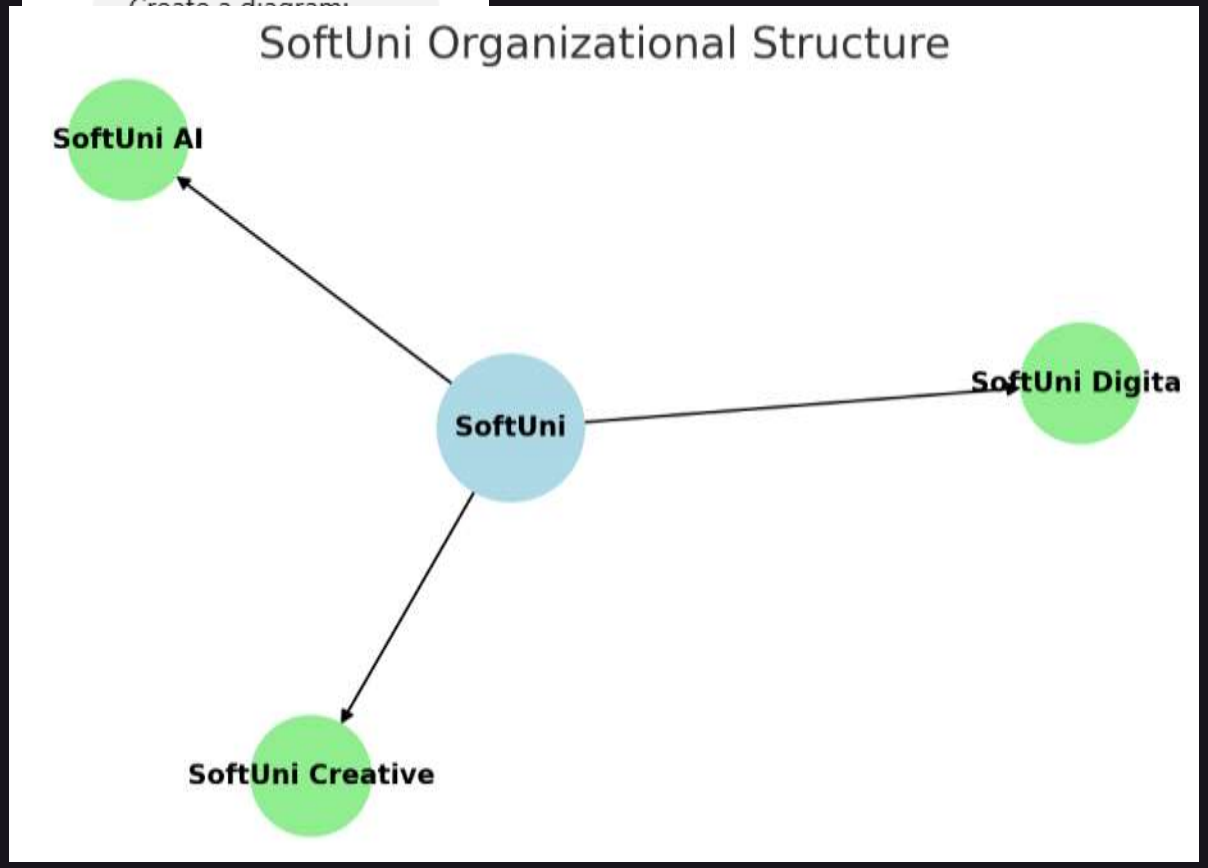
Yes, this topic is highly significant today for several reasons:

1. **Renewable Energy Education**
   - With growing concerns about **climate change**, **fossil fuel** understanding wind power is crucial.
   - This experiment helps students grasp **how wind energy is**
2. **Hands-on STEM Learning**
   - Provides practical application of **physics, engineering, and**
   - Encourages **innovation** in turbine designs, which is relevar **technology**.

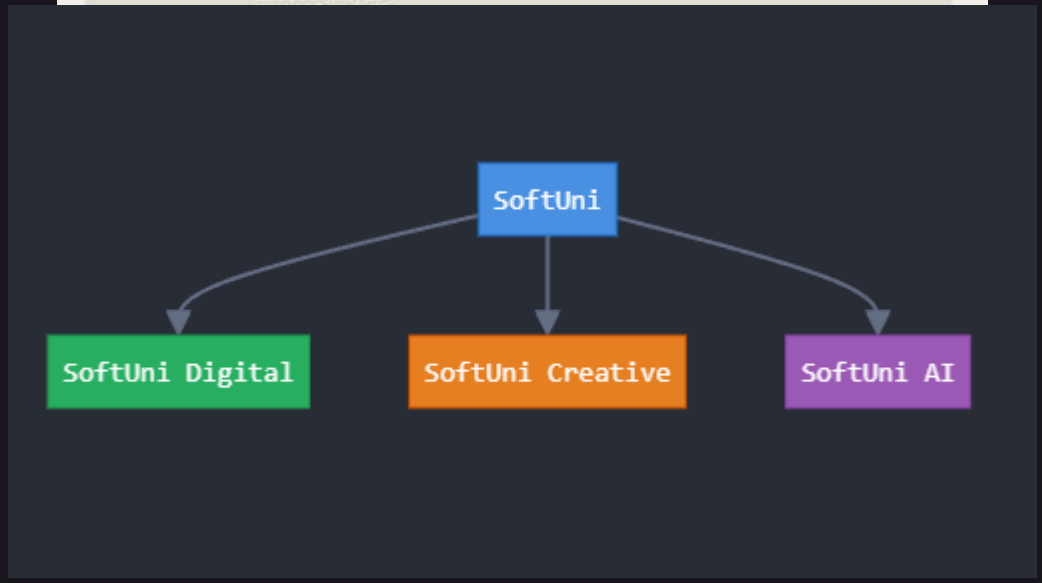
https://chatgpt.com/share/67b31a1e-0304-8002-bc42-659cf4482262

Drawing a Diagram with AI

- **ChatGPT** and **Claude** can draw diagrams:







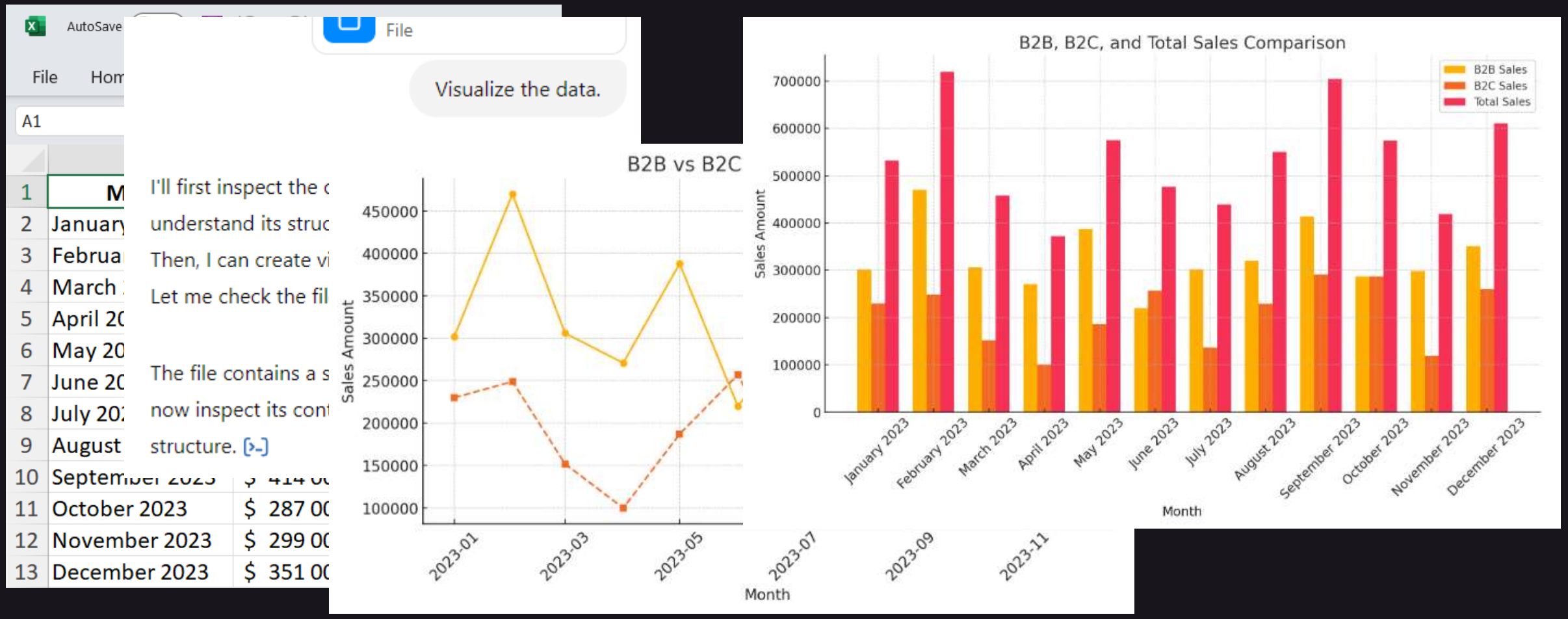
https://chatgpt.com/share/67bc6ab7-2398-8006-9cc9-a5d7c4b978c6

https://claude.site/artifacts/df0818c4-6ae3-411f-9438-c5f559eac930

Visualization of Excel Data

SoftUni AI

- ChatGPT can analyze and visualize data:



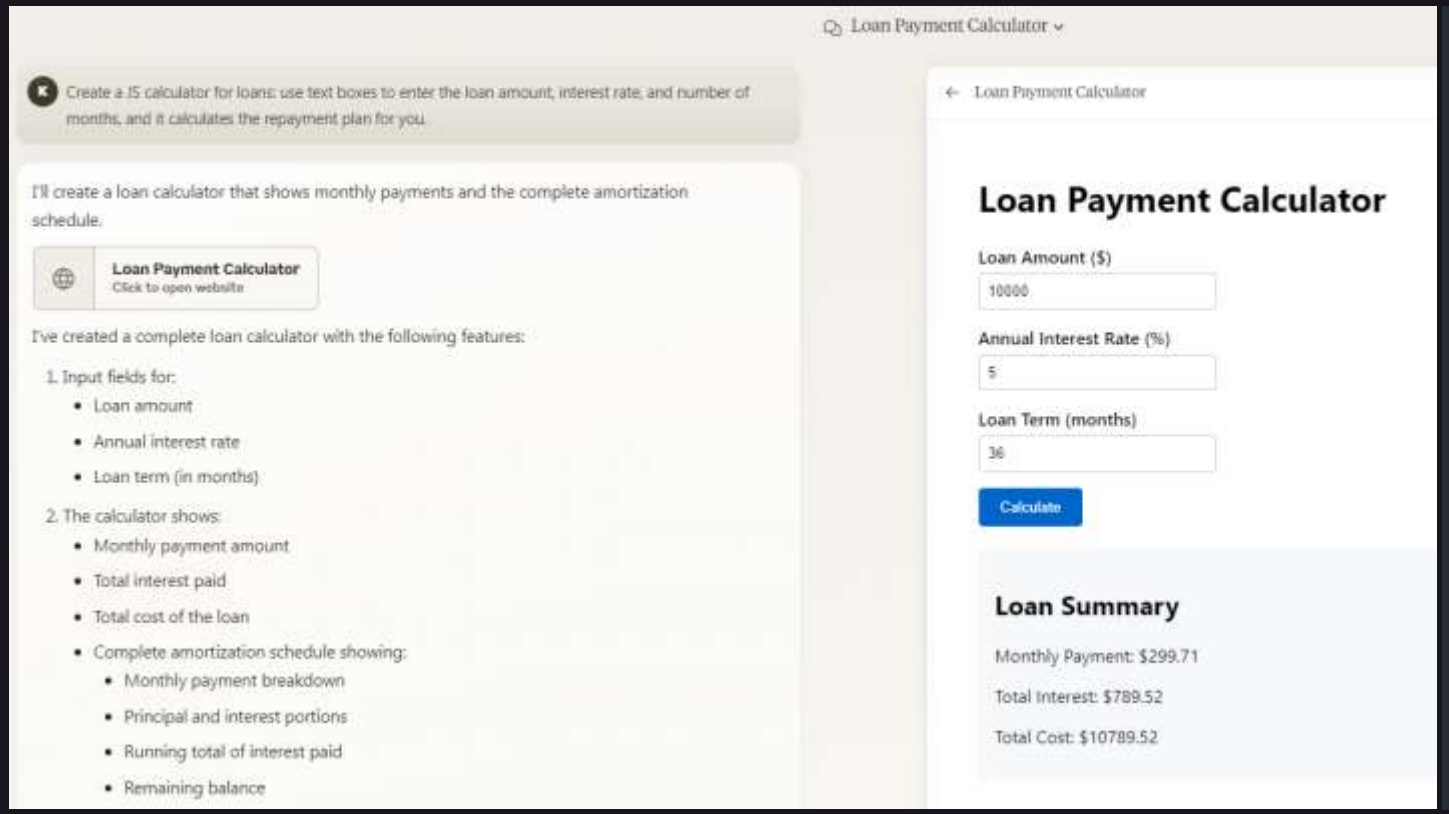
1	M	
2	January	
3	Februa	
4	March	
5	April 20	
6	May 20	
7	June 20	
8	July 20	
9	August	
10	Septem	
11	October 2023	$ 287 0
12	November 2023	$ 299 0
13	December 2023	$ 351 0

https://chatgpt.com/share/67b32152-8c60-8002-82c8-43f2aa272f37

Writing Code with Claude

SoftUni AI

```
Create a JS calculator for loans: use text boxes to enter the loan amount, interest rate, and number of months, and it calculates the repayment plan for you.
```




```
  <div class="input-group">
    <label for="loanTerm">Loan Term (months)</label>
    <input type="number" id="loanTerm" value="36" min="1" step="1">
  </div>

  <button onclick="calculateLoan()">Calculate</button>

  <div class="results">
    <div class="summary" id="summary"></div>
    <div id="schedule"></div>
  </div>
</div>

<script>
  function calculateLoan() {
    const loanAmount = parseFloat(document.getElementById('loanAmount').val
    const annualRate = parseFloat(document.getElementById('interestRate').v
    const months = parseInt(document.getElementById('loanTerm').value);

    // Calculate monthly rate and payment
    const monthlyRate = annualRate / 1200; // Convert annual rate to monthl
    const monthlyPayment = loanAmount * monthlyRate * Math.pow(1 + monthlyR
                           / (Math.pow(1 + monthlyRate, months) - 1);
```

https://claude.site/artifacts/03dd4867-ba41-4c02-946e-eea2762e6dc8

Context Size

Limited Context Size and How It Affects AI Chatbots



What is "Context Size"?

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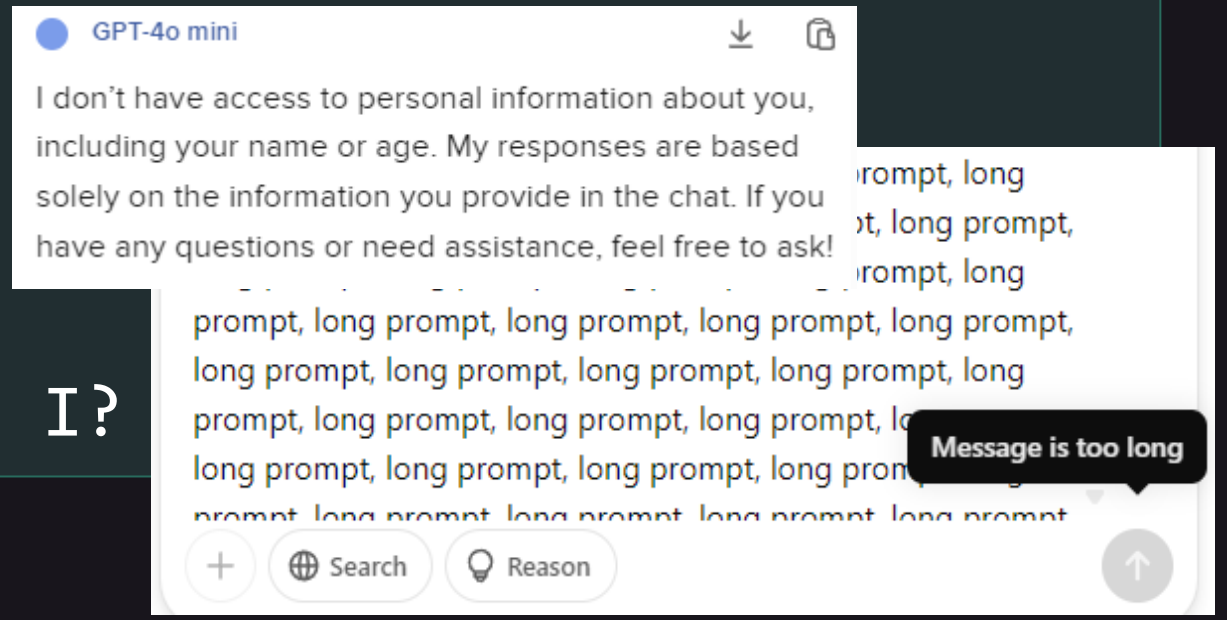
- Context window size
  - The amount of text (in tokens) that is processed by LLM when generating responses (1 token == 0.75 words)
- Longer context → can analyze more extensive input (for example, summarize more extensive books)
- Shorter context → leads to amnesia (forgetting) during longer sessions or more substantial input
- ChatGPT-4 has a context of 8128 tokens (6,000 words)
- ChatGPT-4o has a context of 128k tokens (96,000 words)

Short Context Amnesia

SoftUni AI

- With an insufficiently long context size, LLM models "forget" the beginning of the conversation:

```
My name is Ivan and I'm 25. I work as a "customer
relations specialist" at SoftUni. I respond to email
inquiries like these:
---
{10,000 email messages}
---
What is my name? How old am I?
```


GPT-4o mini
I don't have access to personal information about you, including your name or age. My responses are based solely on the information you provide in the chat. If you have any questions or need assistance, feel free to ask!
prompt, long prompt, long prompt, long

AI Chat Models Have a Limited Context Size!

SoftUni AI

- Large language models have a limited context window size

Model	Tokens	Words	Pages
ChatGPT-4o	128K	96 000	192
Claude 3.7 Sonnet	200K	150 000	300
Gemini 2.5 Pro	2M	1 500 000	3000
Copilot	128K	96 000	192
Llama 3.1 405B Instruct	128K	96 000	192
DeepSeek-V3	128K	96 000	192

Conclusion

SoftUni AI

- **Prompt engineering** - the art of writing effective queries to AI tools
- Prompt structure
  - Context, task/question, input conditions, output format, additional instructions (role, style, tone)
- **Example prompts** – creating a plan, writing an email, clarifying questions, tables, diagrams, …
- **Context size** – how much information can be processed at once (ChatGPT 4o → 128k tokens)



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